



# IdeaVault

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Complete Codebase Architecture Documentation

MERN Stack Application — JavaScript

Generated: July 2026

Version 1.0

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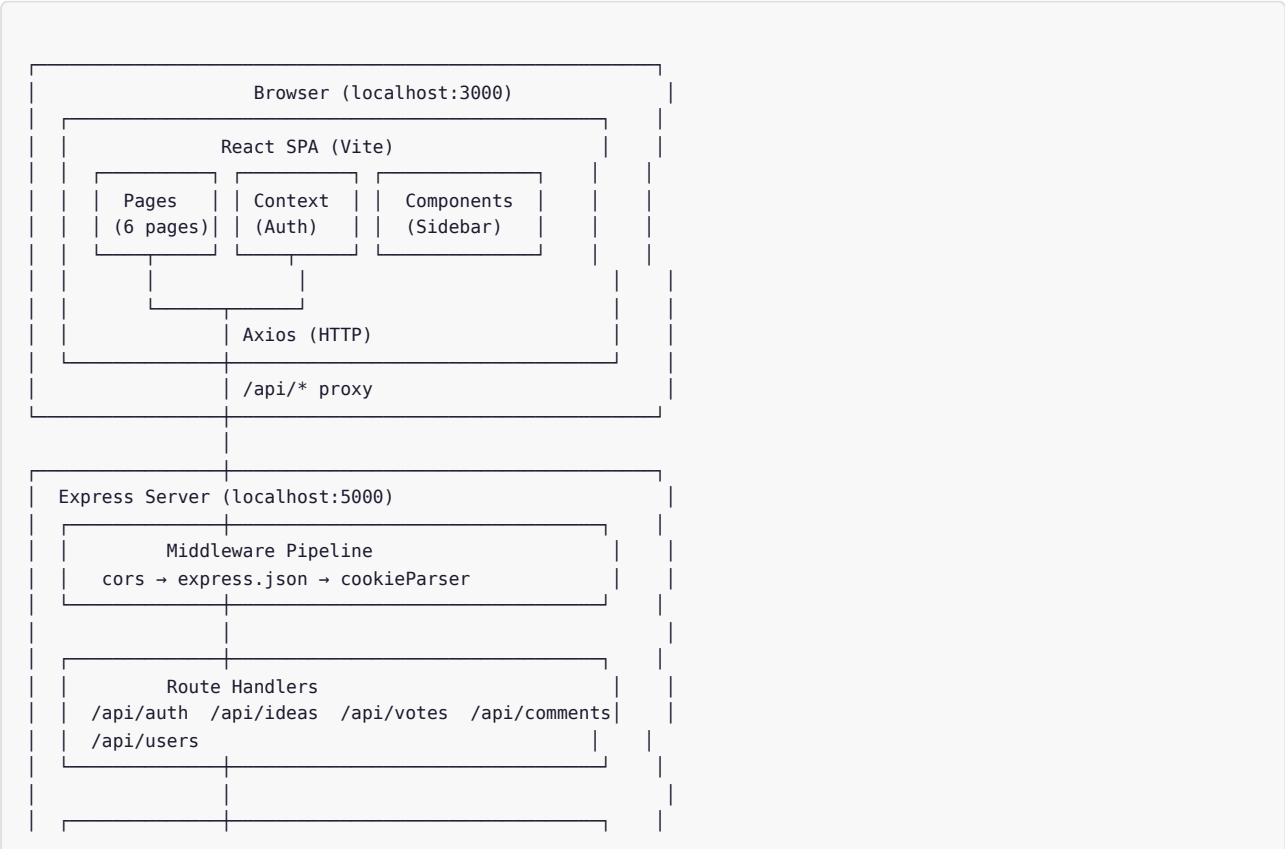
# 1. Project Overview & Tech Stack

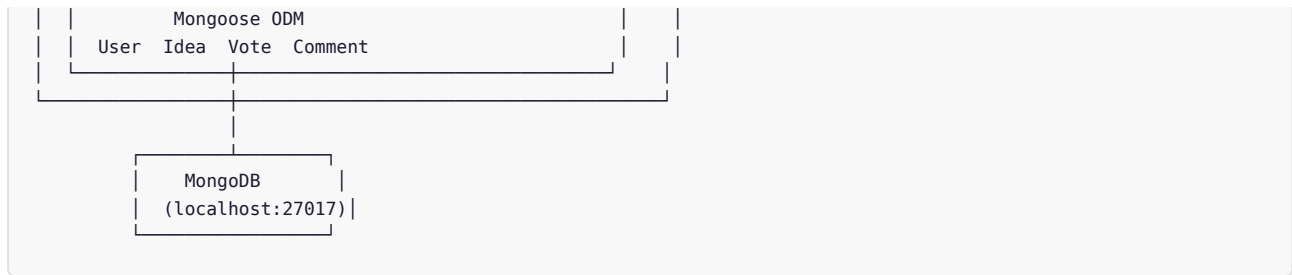
IdeaVault is a MERN stack application where users post ideas publicly and the community votes them good or bad. It uses a monorepo structure with a root `package.json` orchestrating both server and client via `concurrently`.

## Technology Stack

| Layer       | Technology              | Version | Purpose                        |
|-------------|-------------------------|---------|--------------------------------|
| Runtime     | Node.js                 | 18+     | JavaScript runtime             |
| Server      | Express                 | ^4.21.0 | HTTP framework                 |
| Database    | MongoDB + Mongoose      | ^8.6.0  | NoSQL database + ODM           |
| Auth        | jsonwebtoken + bcryptjs | —       | JWT cookies + password hashing |
| Client      | React                   | ^19.2.7 | UI library                     |
| Routing     | react-router-dom        | ^7.18.1 | Client-side routing            |
| Build       | Vite                    | ^5.4.21 | Dev server + bundler           |
| HTTP Client | Axios                   | ^1.18.1 | API requests                   |
| Icons       | lucide-react            | ^1.24.0 | Icon library                   |
| Testing     | node:test + supertest   | ^7.0.0  | Unit & integration tests       |
| Linting     | oxlint                  | ^1.71.0 | Fast JS/React linter           |

## Architecture Diagram





## 2. Complete File Tree

```

idea-vault/
├── package.json          ← Root monorepo orchestrator
├── .gitignore            ← Ignores node_modules/, .env, dist/
├── CONTEXT.md           ← Domain glossary (7 terms)
├── docs/
│   ├── PRD.md           ← Product requirements document
│   └── adr/
│       ├── 0001-binary-voting-with-change.md
│       └── 0002-author-controlled-status.md
├── server/              ← Express.js backend
│   ├── .env             ← MONGO_URI, JWT_SECRET, PORT
│   ├── package.json     ← ESM, server dependencies
│   ├── index.js         ← Entry point: connects DB, starts server
│   ├── seed.js          ← Seeds 3 users, 10 ideas, votes, comments
│   └── src/
│       ├── app.js       ← Express app, middleware, route mounting
│       ├── health.test.js ← Health endpoint test
│       └── auth/
│           ├── user.model.js ← User schema (name, email, password)
│           ├── auth.middleware.js ← JWT verification middleware
│           ├── auth.routes.js ← /register, /login, /me, /logout
│           ├── auth.service.js ← JWT generation, cookie helpers
│           └── auth.test.js ← Auth endpoint tests (7 tests)
│       ├── ideas/
│           ├── idea.model.js ← Idea schema (title, desc, status, author)
│           ├── idea.routes.js ← CRUD for ideas + status update
│           ├── idea.test.js ← Idea endpoint tests (7 tests)
│           ├── score.js ← computeScore() function
│           └── score.test.js ← Score computation tests (5 tests)
│       ├── votes/
│           ├── vote.model.js ← Vote schema (idea, user, value)
│           ├── vote.routes.js ← Cast/change/remove votes
│           └── vote.test.js ← Vote endpoint tests (6 tests)
│       ├── comments/
│           ├── comment.model.js ← Comment schema (text, idea, author)
│           ├── comment.routes.js ← CRUD for comments
│           └── comment.test.js ← Comment endpoint tests (6 tests)
│       └── users/
│           └── user.routes.js ← GET /users/:id/ideas
├── client/              ← React frontend
│   ├── package.json     ← ESM, client dependencies
│   ├── vite.config.js   ← Port 3000, proxy /api to :5000
│   ├── index.html       ← HTML shell with #root
│   ├── public/
│   │   ├── favicon.svg ← Vite lightning bolt logo
│   │   └── icons.svg ← SVG sprite (UNUSED)
│   └── src/
│       ├── main.jsx     ← React entrypoint
│       ├── App.jsx      ← Router, sidebar, ProtectedRoute
│       ├── index.css    ← ALL application styles (769 lines)
│       ├── App.css      ← Empty file
│       └── context/
│           ├── AuthContext.jsx ← Auth state, login/register/logout
│           └── pages/
│               ├── Home.jsx ← Feed with filter/search/sort
│               ├── Login.jsx ← Login form
│               ├── Register.jsx ← Registration form
│               ├── CreateIdea.jsx ← New idea form
│               ├── IdeaDetail.jsx ← Idea view + voting + comments
│               └── Profile.jsx ← User profile + their ideas

```

### 3. Configuration Files

**package.json** ROOT

Monorepo orchestrator using `concurrently` to run server and client simultaneously.

| Script                      | Command                                                     | What It Does                      |
|-----------------------------|-------------------------------------------------------------|-----------------------------------|
| <code>npm run dev</code>    | <code>concurrently "npm run server" "npm run client"</code> | Starts both server and client     |
| <code>npm run server</code> | <code>npm run dev --prefix server</code>                    | Starts Express with file watching |
| <code>npm run client</code> | <code>npm run dev --prefix client</code>                    | Starts Vite dev server            |
| <code>npm run seed</code>   | <code>npm run seed --prefix server</code>                   | Seeds database with sample data   |
| <code>npm test</code>       | <code>npm test --prefix server</code>                       | Runs all server tests             |

**server/package.json** SERVER

ESM module ( `"type": "module"` ). Key dependencies: `express`, `mongoose`, `bcryptjs`, `jsonwebtoken`, `cookie-parser`, `cors`, `dotenv`.

| Script             | Command                                                        |
|--------------------|----------------------------------------------------------------|
| <code>dev</code>   | <code>node --watch index.js</code> (Node native file watching) |
| <code>start</code> | <code>node index.js</code>                                     |
| <code>seed</code>  | <code>node seed.js</code>                                      |
| <code>test</code>  | <code>node --test src/**/*.test.js</code>                      |

**client/package.json** CLIENT

ESM module. Key dependencies: `react`, `react-dom`, `react-router-dom`, `axios`, `lucide-react`. Dev: `vite`, `@vitejs/plugin-react`, `oxlint`.

**client/vite.config.js** CONFIG

Configures the Vite dev server:

- Dev server runs on **port 3000**
- Proxy: all `/api` requests are forwarded to `http://localhost:5000`
- React plugin enabled via `@vitejs/plugin-react`

**server/.env** ENV

```
MONGO_URI=mongodb://localhost:27017/idea-vault
JWT_SECRET=idea-vault-dev-secret-key
PORT=5000
```

**client/.oxlintrc.json** LINT

Configures oxlint with React plugin. Rules: `react/rules-of-hooks` (error), `react/only-export-components` (warn).

## 4. Server Architecture

### Startup Sequence

```
server/index.js
1. Loads dotenv (configures process.env from .env)
2. Imports app.js (the Express application)
3. Connects to MongoDB via mongoose.connect(MONGO_URI)
4. Starts listening on PORT (default 5000)
```

### Express Middleware Pipeline (app.js)

Requests pass through these middleware in order:

| # | Middleware                                                                | Purpose                                                             |
|---|---------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1 | <code>cors({ origin: "http://localhost:3000", credentials: true })</code> | Allows cross-origin requests from the React dev server with cookies |
| 2 | <code>express.json()</code>                                               | Parses JSON request bodies                                          |
| 3 | <code>cookieParser()</code>                                               | Parses cookies from the Cookie header                               |
| 4 | Route handlers                                                            | See route mounting below                                            |
| 5 | Error handler                                                             | Catches errors, returns <code>{ error: { message } }</code>         |

### Route Mounting

| Mount Point                  | Route File                 | What It Handles                       |
|------------------------------|----------------------------|---------------------------------------|
| <code>GET /api/health</code> | app.js (inline)            | Returns <code>{ status: "ok" }</code> |
| <code>/api/auth/*</code>     | auth/auth.routes.js        | Register, login, me, logout           |
| <code>/api/users/*</code>    | users/user.routes.js       | User ideas endpoint                   |
| <code>/api/ideas/*</code>    | ideas/idea.routes.js       | CRUD for ideas + status               |
| <code>/api/ideas/*</code>    | votes/vote.routes.js       | Vote operations                       |
| <code>/api/ideas/*</code>    | comments/comment.routes.js | Comment operations                    |

**Note:** Both `vote.routes.js` and `comment.routes.js` are mounted on `/api/ideas` because they handle sub-resources of ideas (e.g., `/api/ideas/:id/votes`).

# 5. Server Files — Detailed Breakdown

## 5.1 Entry Point & App Setup

server/index.js ENTRY POINT

**Purpose:** Application entry point. Loads environment variables, connects to MongoDB, starts Express.

**Imports:** dotenv, ./src/app.js, mongoose (dynamic import)

**Exports:** None (executable script)

**Functions:** None (top-level async code)

server/src/app.js APP SETUP

**Purpose:** Creates and configures the Express application with all middleware and routes.

**Imports:** express, cors, cookie-parser, and all 5 route modules

**Exports:** app (default export) — the configured Express app instance

**Functions:** None (setup only)

## 5.2 Authentication Module (server/src/auth/)

user.model.js MODEL

**Purpose:** Defines the User MongoDB schema and model.

**Schema Fields:**

| Field     | Type   | Constraints                          | Notes                            |
|-----------|--------|--------------------------------------|----------------------------------|
| name      | String | required                             | User's display name              |
| email     | String | required, unique, lowercase          | Login identifier                 |
| password  | String | required, <code>select: false</code> | Excluded from queries by default |
| createdAt | Date   | default: Date.now                    | Account creation timestamp       |

**Pre-save hook:** Hashes password with bcrypt (10 rounds) when modified.

**Instance methods:** `comparePassword(candidate)` — uses `bcrypt.compare` to verify password.

auth.service.js SERVICE

**Purpose:** JWT and cookie helper functions.

**Exports (named):**

| Function                                | Parameters        | Returns    | Description                                                       |
|-----------------------------------------|-------------------|------------|-------------------------------------------------------------------|
| <code>generateToken(userId)</code>      | userId (ObjectId) | JWT string | Signs JWT with <code>{ id: userId }</code> , expires in 7 days    |
| <code>setTokenCookie(res, token)</code> | res, token        | void       | Sets httpOnly cookie named "token", maxAge 7 days, sameSite "lax" |



|                                    |                  |                   |                                                           |
|------------------------------------|------------------|-------------------|-----------------------------------------------------------|
| <code>clearTokenCookie(res)</code> | <code>res</code> | <code>void</code> | Clears "token" cookie by setting <code>maxAge</code> to 0 |
|------------------------------------|------------------|-------------------|-----------------------------------------------------------|

`auth.middleware.js` MIDDLEWARE

**Purpose:** JWT verification middleware for protected routes.

**Exports:** `auth` (default) — Express middleware function

**Behavior:** Extracts `token` from `req.cookies.token`, verifies JWT, sets `req.userId = decoded.id`. Returns 401 if missing or invalid.

`auth.routes.js` ROUTES

**Purpose:** Handles all authentication endpoints.

**Exports:** `router` (default)

| Method | Path      | Auth              | Body                      | Response                             |
|--------|-----------|-------------------|---------------------------|--------------------------------------|
| POST   | /register | No                | { name, email, password } | { token, user: { id, name, email } } |
| POST   | /login    | No                | { email, password }       | { token, user: { id, name, email } } |
| GET    | /me       | No (reads cookie) | —                         | { user: { id, name, email } }        |
| POST   | /logout   | No                | —                         | { message: "Logged out" }            |

5.3 Ideas Module (server/src/ideas/)

`idea.model.js` MODEL

**Schema Fields:**

| Field       | Type     | Constraints              | Default  |
|-------------|----------|--------------------------|----------|
| title       | String   | required                 | —        |
| description | String   | required                 | —        |
| status      | String   | enum: ["open", "closed"] | "open"   |
| author      | ObjectId | ref: "User", required    | —        |
| createdAt   | Date     | —                        | Date.now |
| updatedAt   | Date     | —                        | Date.now |

**Pre-save hook:** Sets `updatedAt` to `new Date()`.

`score.js` UTILITY

**Purpose:** Pure function to compute vote scores.

**Exports:** `computeScore(votes)` (named)

**Logic:** Filters votes for "good" and "bad", computes percentage as `Math.round((good / total) * 100)` (0 if no votes).

**Returns:** { good: number, bad: number, percentage: number }

**Called by:** `idea.routes.js`, `vote.routes.js`, `user.routes.js`

**idea.routes.js** ROUTES

| Method | Path | Auth | Description                                                                                                                                           |
|--------|------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET    | /    | No   | List ideas. Query params: <code>sort</code> (newest popular), <code>status</code> (open closed all), <code>search</code> (regex on title+description) |
| GET    | /:id | No   | Get single idea by ID with computed score                                                                                                             |
| POST   | /    | Yes  | Create idea (title, description)                                                                                                                      |
| PATCH  | /:id | Yes  | Update idea status (author only)                                                                                                                      |

## 5.4 Votes Module (server/src/votes/)

**vote.model.js** MODEL

| Field | Type     | Constraints                     |
|-------|----------|---------------------------------|
| idea  | ObjectId | ref: "Idea", required           |
| user  | ObjectId | ref: "User", required           |
| value | String   | enum: ["good", "bad"], required |

**Index:** Compound unique index on `{ idea: 1, user: 1 }` — enforces one vote per user per idea.

**vote.routes.js** ROUTES

| Method | Path          | Auth | Description                                                                            |
|--------|---------------|------|----------------------------------------------------------------------------------------|
| POST   | /:id/votes    | Yes  | Cast or change vote. Validates: idea exists, is open, user is not author. Uses upsert. |
| GET    | /:id/votes/me | Yes  | Get current user's vote on this idea                                                   |
| DELETE | /:id/votes    | Yes  | Remove user's vote                                                                     |

## 5.5 Comments Module (server/src/comments/)

**comment.model.js** MODEL

| Field     | Type     | Constraints           |
|-----------|----------|-----------------------|
| text      | String   | required              |
| idea      | ObjectId | ref: "Idea", required |
| author    | ObjectId | ref: "User", required |
| createdAt | Date     | default: Date.now     |
| updatedAt | Date     | default: Date.now     |

**Pre-save hook:** Sets `updatedAt` to `new Date()`.

**comment.routes.js** ROUTES

| Method | Path                         | Auth | Description                                                                       |
|--------|------------------------------|------|-----------------------------------------------------------------------------------|
| GET    | /:id/comments                | No   | List comments for an idea. Populates author name. Sorted by createdAt descending. |
| POST   | /:id/comments                | Yes  | Add comment. Body: { text }.                                                      |
| PATCH  | /:ideaId/comments/:commentId | Yes  | Edit comment (comment author only). Body: { text }.                               |
| DELETE | /:ideaId/comments/:commentId | Yes  | Delete comment (idea author only, not comment author).                            |

**Note:** DELETE uses dynamic import of Idea model rather than static import at top of file.

## 5.6 Users Module (server/src/users/)

user.routes.js ROUTES

| Method | Path             | Auth | Description                                           |
|--------|------------------|------|-------------------------------------------------------|
| GET    | /users/:id/ideas | No   | Get all ideas by a specific user with computed scores |

## 5.7 Seed Script

server/seed.js SCRIPT

**Purpose:** Clears all collections, then creates sample data for development.

**Creates:**

- 3 users: alice@example.com, bob@example.com, carol@example.com (all password: password123)
- 10 ideas with varied titles and descriptions
- Votes: each non-author voter votes ~70% of the time
- Comments: 1-3 comments per idea

## 6. API Route Reference

Complete table of all 17 API endpoints. Base URL: `http://localhost:5000`

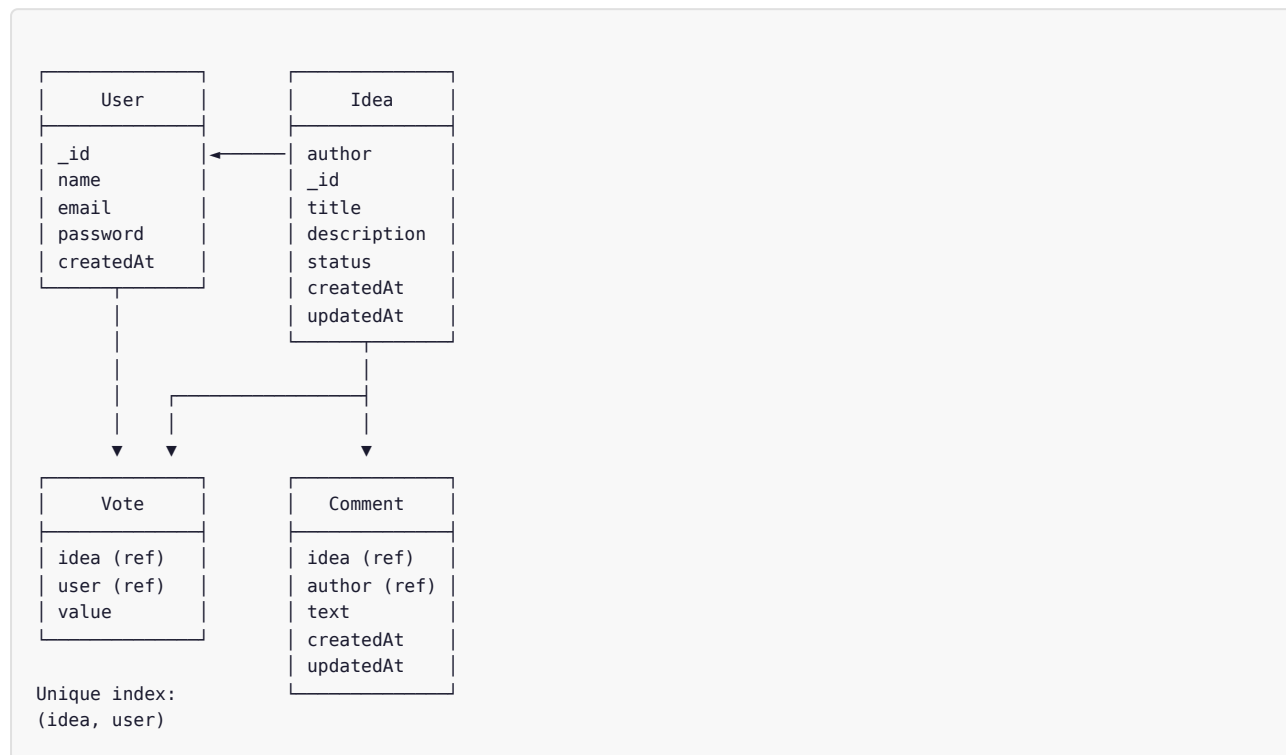
| #  | Method | Path                                   | Auth   | Handler File      | Request Body                | Response                             |
|----|--------|----------------------------------------|--------|-------------------|-----------------------------|--------------------------------------|
| 1  | GET    | /api/health                            | No     | app.js (inline)   | —                           | { status: "ok" }                     |
| 2  | POST   | /api/auth/register                     | No     | auth.routes.js    | { name, email, password }   | { token, user: { id, name, email } } |
| 3  | POST   | /api/auth/login                        | No     | auth.routes.js    | { email, password }         | { token, user: { id, name, email } } |
| 4  | GET    | /api/auth/me                           | Cookie | auth.routes.js    | —                           | { user: { id, name, email } }        |
| 5  | POST   | /api/auth/logout                       | No     | auth.routes.js    | —                           | { message: "Logged out" }            |
| 6  | GET    | /api/ideas                             | No     | idea.routes.js    | —                           | { ideas[], total }                   |
| 7  | GET    | /api/ideas/:id                         | No     | idea.routes.js    | —                           | { idea }                             |
| 8  | POST   | /api/ideas                             | JWT    | idea.routes.js    | { title, description }      | { idea }                             |
| 9  | PATCH  | /api/ideas/:id                         | JWT    | idea.routes.js    | { status: "open" "closed" } | { idea }                             |
| 10 | POST   | /api/ideas/:id/votes                   | JWT    | vote.routes.js    | { value: "good" "bad" }     | { score }                            |
| 11 | GET    | /api/ideas/:id/votes/me                | JWT    | vote.routes.js    | —                           | { vote }   null                      |
| 12 | DELETE | /api/ideas/:id/votes                   | JWT    | vote.routes.js    | —                           | { score }                            |
| 13 | GET    | /api/ideas/:id/comments                | No     | comment.routes.js | —                           | { comments[] }                       |
| 14 | POST   | /api/ideas/:id/comments                | JWT    | comment.routes.js | { text }                    | { comment }                          |
| 15 | PATCH  | /api/ideas/:ideald/comments/:commentId | JWT    | comment.routes.js | { text }                    | { comment }                          |
| 16 | DELETE | /api/ideas/:ideald/comments/:commentId | JWT    | comment.routes.js | —                           | { success: true }                    |
| 17 | GET    | /api/users/:id/ideas                   | No     | user.routes.js    | —                           | { ideas[], total }                   |

### Query Parameters for GET /api/ideas

| Param  | Values            | Default | Effect                                                   |
|--------|-------------------|---------|----------------------------------------------------------|
| sort   | newest, popular   | newest  | Sort by creation date or vote count                      |
| status | open, closed, all | all     | Filter by idea status                                    |
| search | string            | —       | Regex search on title and description (case-insensitive) |

## 7. MongoDB Models

### Entity Relationship Diagram



### Score Computation

The `computeScore(votes)` function in `score.js` is called at read time (never stored):

```
Input: votes = [{ value: "good" }, { value: "bad" }, { value: "good" }]
Output: { good: 2, bad: 1, percentage: 67 }
```

Logic:

```
good = votes.filter(v => v.value === "good").length
bad = votes.filter(v => v.value === "bad").length
total = good + bad
percentage = total === 0 ? 0 : Math.round((good / total) * 100)
```

## 8. Client Architecture

### Startup Sequence

```
client/src/main.jsx
1. Imports index.css (global styles)
2. Renders <App /> inside <StrictMode>
3. Mounts into #root div in index.html
```

### Component Tree

```
<BrowserRouter>
├── <AuthProvider>      ← context/AuthContext.jsx
│   └── <AppLayout>      ← App.jsx
│       ├── <Sidebar>    ← App.jsx (shown when logged in)
│           ├── Link to /      (Home icon)
│           ├── Link to /ideas/new (PlusCircle icon)
│           ├── Link to /profile/:id (User icon)
│           ├── Button logout   (LogOut icon)
│           └── User info div
│       └── <Routes>
│           ├── /register      → <Register />
│           ├── /login        → <Login />
│           ├── /             → <ProtectedRoute> → <Home />
│           ├── /ideas/new    → <ProtectedRoute> → <CreateIdea />
│           ├── /ideas/:id    → <ProtectedRoute> → <IdeaDetail />
│           └── /profile/:id  → <ProtectedRoute> → <Profile />
```

### Route Table

| URL Pattern  | Component      | Auth Required | Description                   |
|--------------|----------------|---------------|-------------------------------|
| /register    | Register.jsx   | No            | Registration form             |
| /login       | Login.jsx      | No            | Login form                    |
| /            | Home.jsx       | Yes           | Idea feed with filters        |
| /ideas/new   | CreateIdea.jsx | Yes           | Create new idea form          |
| /ideas/:id   | IdeaDetail.jsx | Yes           | View idea, vote, comment      |
| /profile/:id | Profile.jsx    | Yes           | User profile with their ideas |

### Auth Flow

```
AuthContext.jsx provides: { user, loading, login, register, logout }
```

On mount:

```
→ GET /api/auth/me (reads cookie, restores session)
→ user = { id, name, email } or null
```

```
login(email, password):
```

```
→ POST /api/auth/login
→ Sets user state
```

```
register(name, email, password):
```

```
→ POST /api/auth/register
```

→ Sets user state

logout():

→ POST /api/auth/logout

→ Clears user state, redirects to /login

## 9. Client Files — Detailed Breakdown

### main.jsx ENTRY

**Purpose:** React entrypoint. Imports global CSS and renders App.

**Imports:** StrictMode (react), createRoot (react-dom), index.css, App.jsx

**Exports:** None

### App.jsx ROOT + ROUTER

**Purpose:** Root component. Defines routing, sidebar, and ProtectedRoute.

**Components defined:**

- ProtectedRoute({ children }) — redirects to /login if not authenticated
- App() — wraps everything in BrowserRouter → AuthProvider → AppLayout
- AppLayout() — renders Sidebar (if logged in) + Routes
- Sidebar() — navigation with icons and user info

**Icons used:** Home (24), Lightbulb (28), PlusCircle (24), User (24), LogOut (24)

**CSS classes:** loading, app, sidebar, sidebar-logo, sidebar-nav, active, sidebar-user, sidebar-user-avatar, sidebar-user-info, sidebar-user-name, sidebar-user-email, main-content

### context/AuthContext.jsx CONTEXT

**Purpose:** Provides authentication state and methods to all components.

**Exports:** AuthProvider (component), useAuth (hook)

**State:** user (null | { id, name, email }), loading (boolean)

**API calls made:**

- On mount: GET /api/auth/me — restores session from cookie
- register(name, email, password) : POST /api/auth/register
- login(email, password) : POST /api/auth/login
- logout() : POST /api/auth/logout

### pages/Home.jsx PAGE

**Route:** / (protected)

**State:** ideas, loading, sort (newest|popular), status (all|open|closed), search

**API calls:** GET /api/ideas?sort=...&status=...&search=... (debounced 300ms)

**Icons:** ThumbsUp (14), ThumbsDown (14)

**CSS classes:** main-header, filter-bar, status-tabs, tab, active, filter-group, filter-group search, loading, empty-state, idea-list, idea-card, idea-card-author, idea-card-header, status-badge, idea-excerpt, idea-card-footer

### pages/Login.jsx PAGE

**Route:** /login (unprotected)

**State:** form ({ email, password }), error

**API calls:** Uses login() from AuthContext → POST /api/auth/login

**Icons:** Lightbulb (48)



**CSS classes:** auth-page, subtitle, error, switch-link

### pages/Register.jsx PAGE

**Route:** /register (unprotected)

**State:** form ({ name, email, password }), error

**API calls:** Uses `register()` from `AuthContext` → `POST /api/auth/register`

**Icons:** Lightbulb (48)

**CSS classes:** auth-page, subtitle, error, switch-link

### pages/CreateIdea.jsx PAGE

**Route:** /ideas/new (protected)

**State:** form ({ title, description }), error, submitting

**API calls:** `POST /api/ideas` with { title, description }, then navigates to `/ideas/:id`

**Icons:** ArrowLeft (20)

**CSS classes:** main-header, error, create-page, create-form, create-form-actions, btn-primary

### pages/IdeaDetail.jsx PAGE (most complex)

**Route:** /ideas/:id (protected)

**State:** idea, loading, userVote, comments, commentText, editingId, editText, submitting

**API calls (9 total):**

- `GET /api/ideas/${id}` — fetch idea
- `GET /api/ideas/${id}/comments` — fetch comments
- `GET /api/ideas/${id}/votes/me` — fetch user's vote
- `POST /api/ideas/${id}/votes` — cast vote
- `DELETE /api/ideas/${id}/votes` — remove vote
- `POST /api/ideas/${id}/comments` — add comment
- `PATCH /api/ideas/${id}/comments/${cid}` — edit comment
- `DELETE /api/ideas/${id}/comments/${cid}` — delete comment
- `PATCH /api/ideas/${id}` — toggle status

**Icons:** ThumbsUp (16), ThumbsDown (16), MessageCircle (20), Trash2 (14), Edit2 (14), ArrowLeft (20)

**Computed values:** `isAuthor = user.id === idea.author._id`, `canVote = user && !isAuthor && idea.status === "open"`

### pages/Profile.jsx PAGE

**Route:** /profile/:id (protected)

**State:** ideas, loading, profileUser

**API calls:** `GET /api/users/${id}/ideas`

**Icons:** ThumbsUp (14), ThumbsDown (14), ArrowLeft (20)

**CSS classes:** loading, profile-page, main-header, profile-header, profile-avatar, profile-name, profile-stats, empty-state, idea-list, idea-card, idea-card-header, status-badge, idea-card-footer

## 10. CSS → Component Mapping

Every CSS class in `index.css` mapped to which component(s) use it:

| CSS Class                             | Used By Component(s)                  |
|---------------------------------------|---------------------------------------|
| <code>.app</code>                     | App.jsx                               |
| <code>.sidebar</code>                 | App.jsx (Sidebar)                     |
| <code>.sidebar-logo</code>            | App.jsx (Sidebar)                     |
| <code>.sidebar-logo a</code>          | App.jsx (Sidebar)                     |
| <code>.sidebar-nav</code>             | App.jsx (Sidebar)                     |
| <code>.sidebar-nav a</code>           | App.jsx (Sidebar)                     |
| <code>.sidebar-nav button</code>      | App.jsx (Sidebar)                     |
| <code>.sidebar-nav a.active</code>    | App.jsx (Sidebar)                     |
| <code>.sidebar-user</code>            | App.jsx (Sidebar)                     |
| <code>.sidebar-user-avatar</code>     | App.jsx (Sidebar)                     |
| <code>.sidebar-user-info</code>       | App.jsx (Sidebar)                     |
| <code>.sidebar-user-name</code>       | App.jsx (Sidebar)                     |
| <code>.sidebar-user-email</code>      | App.jsx (Sidebar)                     |
| <code>.main-content</code>            | App.jsx                               |
| <code>.main-header</code>             | Home, CreateIdea, IdeaDetail, Profile |
| <code>.main-header h2</code>          | Home, CreateIdea, IdeaDetail, Profile |
| <code>.idea-card</code>               | Home, Profile                         |
| <code>.idea-card-author</code>        | Home                                  |
| <code>.idea-card-header</code>        | Home, Profile                         |
| <code>.idea-card-header h3</code>     | Home, Profile                         |
| <code>.idea-excerpt</code>            | Home                                  |
| <code>.idea-card-footer</code>        | Home, Profile                         |
| <code>.status-badge</code>            | Home, IdeaDetail, Profile             |
| <code>.status-badge.open</code>       | Home, IdeaDetail, Profile             |
| <code>.status-badge.closed</code>     | Home, IdeaDetail, Profile             |
| <code>.idea-detail</code>             | IdeaDetail                            |
| <code>.idea-detail-title</code>       | IdeaDetail                            |
| <code>.idea-detail-description</code> | IdeaDetail                            |
| <code>.idea-detail-score</code>       | IdeaDetail                            |
| <code>.idea-detail-avatar</code>      | IdeaDetail                            |
| <code>.idea-detail-author</code>      | IdeaDetail                            |

|                                         |                 |
|-----------------------------------------|-----------------|
| <code>.vote-buttons</code>              | IdeaDetail      |
| <code>.vote-btn</code>                  | IdeaDetail      |
| <code>.vote-btn.good.active</code>      | IdeaDetail      |
| <code>.vote-btn.bad.active</code>       | IdeaDetail      |
| <code>.comments-section</code>          | IdeaDetail      |
| <code>.comment-form</code>              | IdeaDetail      |
| <code>.comment-form-avatar</code>       | IdeaDetail      |
| <code>.comment-form-input</code>        | IdeaDetail      |
| <code>.comment-form-actions</code>      | IdeaDetail      |
| <code>.comment-list</code>              | IdeaDetail      |
| <code>.comment</code>                   | IdeaDetail      |
| <code>.comment-avatar</code>            | IdeaDetail      |
| <code>.comment-body</code>              | IdeaDetail      |
| <code>.comment-meta</code>              | IdeaDetail      |
| <code>.comment-text</code>              | IdeaDetail      |
| <code>.comment-actions</code>           | IdeaDetail      |
| <code>.comment-edit</code>              | IdeaDetail      |
| <code>.comment-edit-actions</code>      | IdeaDetail      |
| <code>.filter-bar</code>                | Home            |
| <code>.status-tabs</code>               | Home            |
| <code>.tab</code>                       | Home            |
| <code>.tab.active</code>                | Home            |
| <code>.filter-group</code>              | Home            |
| <code>.filter-group.search input</code> | Home            |
| <code>.create-page</code>               | CreateIdea      |
| <code>.create-form</code>               | CreateIdea      |
| <code>.create-form-actions</code>       | CreateIdea      |
| <code>.auth-page</code>                 | Login, Register |
| <code>.auth-page h2</code>              | Login, Register |
| <code>.auth-page .subtitle</code>       | Login, Register |
| <code>.auth-page input</code>           | Login, Register |
| <code>.auth-page button</code>          | Login, Register |
| <code>.auth-page .switch-link</code>    | Login, Register |
| <code>.profile-header</code>            | Profile         |
| <code>.profile-avatar</code>            | Profile         |
| <code>.profile-name</code>              | Profile         |

|                                    |                                         |
|------------------------------------|-----------------------------------------|
| <code>.profile-stats</code>        | Profile                                 |
| <code>.btn-primary</code>          | CreateIdea, IdeaDetail                  |
| <code>.btn-primary:disabled</code> | CreateIdea, IdeaDetail                  |
| <code>.btn-outline</code>          | IdeaDetail                              |
| <code>.loading</code>              | App, Home, IdeaDetail, Profile          |
| <code>.empty-state</code>          | Home, IdeaDetail, Profile               |
| <code>.error</code>                | Login, Register, CreateIdea, IdeaDetail |

## 11. Icon Usage Reference

All icons from the `lucide-react` library:

| Icon          | File(s)                 | Size | Context                   |
|---------------|-------------------------|------|---------------------------|
| Home          | App.jsx                 | 24   | Sidebar navigation link   |
| Lightbulb     | App.jsx                 | 28   | Sidebar logo              |
| Lightbulb     | Login.jsx, Register.jsx | 48   | Auth page header icon     |
| PlusCircle    | App.jsx                 | 24   | Sidebar "New Idea" link   |
| User          | App.jsx                 | 24   | Sidebar "Profile" link    |
| LogOut        | App.jsx                 | 24   | Sidebar logout button     |
| ThumbsUp      | Home.jsx                | 14   | Idea card good vote count |
| ThumbsUp      | IdeaDetail.jsx          | 16   | Vote button               |
| ThumbsUp      | Profile.jsx             | 14   | Idea card good vote count |
| ThumbsDown    | Home.jsx                | 14   | Idea card bad vote count  |
| ThumbsDown    | IdeaDetail.jsx          | 16   | Vote button               |
| ThumbsDown    | Profile.jsx             | 14   | Idea card bad vote count  |
| MessageCircle | IdeaDetail.jsx          | 20   | Comments section header   |
| ArrowLeft     | CreateIdea.jsx          | 20   | Back navigation button    |
| ArrowLeft     | IdeaDetail.jsx          | 20   | Back navigation button    |
| ArrowLeft     | Profile.jsx             | 20   | Back navigation button    |
| Trash2        | IdeaDetail.jsx          | 14   | Delete comment button     |
| Edit2         | IdeaDetail.jsx          | 14   | Edit comment button       |

**Note:** `MessageCircle` is imported in Home.jsx but never rendered in JSX — unused import.

## 12. Frontend ↔ Backend API Mapping

Which frontend file calls which backend endpoint:

| Frontend File   | API Call                                     | Server Endpoint                                  | Server File       |
|-----------------|----------------------------------------------|--------------------------------------------------|-------------------|
| AuthContext.jsx | GET /api/auth/me                             | GET /api/auth/me                                 | auth.routes.js    |
| AuthContext.jsx | POST /api/auth/register                      | POST /api/auth/register                          | auth.routes.js    |
| AuthContext.jsx | POST /api/auth/login                         | POST /api/auth/login                             | auth.routes.js    |
| AuthContext.jsx | POST /api/auth/logout                        | POST /api/auth/logout                            | auth.routes.js    |
| Home.jsx        | GET /api/ideas?<br>sort=&status=&search=     | GET /api/ideas                                   | idea.routes.js    |
| CreateIdea.jsx  | POST /api/ideas                              | POST /api/ideas                                  | idea.routes.js    |
| IdeaDetail.jsx  | GET /api/ideas/\${id}                        | GET /api/ideas/:id                               | idea.routes.js    |
| IdeaDetail.jsx  | GET /api/ideas/\${id}/comments               | GET /api/ideas/:id/comments                      | comment.routes.js |
| IdeaDetail.jsx  | GET /api/ideas/\${id}/votes/me               | GET /api/ideas/:id/votes/me                      | vote.routes.js    |
| IdeaDetail.jsx  | POST /api/ideas/\${id}/votes                 | POST /api/ideas/:id/votes                        | vote.routes.js    |
| IdeaDetail.jsx  | DELETE /api/ideas/\${id}/votes               | DELETE /api/ideas/:id/votes                      | vote.routes.js    |
| IdeaDetail.jsx  | POST /api/ideas/\${id}/comments              | POST /api/ideas/:id/comments                     | comment.routes.js |
| IdeaDetail.jsx  | PATCH<br>/api/ideas/\${id}/comments/\${cid}  | PATCH<br>/api/ideas/:ideaId/comments/:commentId  | comment.routes.js |
| IdeaDetail.jsx  | DELETE<br>/api/ideas/\${id}/comments/\${cid} | DELETE<br>/api/ideas/:ideaId/comments/:commentId | comment.routes.js |
| IdeaDetail.jsx  | PATCH /api/ideas/\${id}                      | PATCH /api/ideas/:id                             | idea.routes.js    |
| Profile.jsx     | GET /api/users/\${id}/ideas                  | GET /api/users/:id/ideas                         | user.routes.js    |

## 13. Import Dependency Graphs

---

### Server Dependency Graph

```

server/index.js
├── dotenv
├── ./src/app.js
└── mongoose (dynamic)

server/src/app.js
├── express
├── cors
├── cookie-parser
├── ./auth/auth.routes.js
├── ./ideas/idea.routes.js
├── ./votes/vote.routes.js
├── ./comments/comment.routes.js
└── ./users/user.routes.js

server/src/auth/auth.routes.js
├── express.Router
├── ./user.model.js
└── ./auth.service.js

server/src/auth/auth.middleware.js
└── jsonwebtoken

server/src/auth/auth.service.js
└── jsonwebtoken

server/src/auth/user.model.js
├── mongoose
└── bcryptjs

server/src/ideas/idea.routes.js
├── express.Router
├── ./idea.model.js
├── ../votes/vote.model.js
├── ../auth/auth.middleware.js
└── ./score.js

server/src/ideas/idea.model.js
└── mongoose

server/src/ideas/score.js
└── (no imports – pure function)

server/src/votes/vote.routes.js
├── express.Router
├── ./vote.model.js
├── ../ideas/idea.model.js
├── ../ideas/score.js
└── ../auth/auth.middleware.js

server/src/votes/vote.model.js
└── mongoose

server/src/comments/comment.routes.js
├── express.Router
├── ./comment.model.js
├── ../auth/auth.middleware.js
└── ../ideas/idea.model.js (dynamic import in DELETE)

server/src/comments/comment.model.js
└── mongoose

```

```
server/src/users/user.routes.js
├─ express.Router
├─ ../ideas/idea.model.js
├─ ../votes/vote.model.js
└─ ../ideas/score.js
```

## Client Dependency Graph

```
client/src/main.jsx
├─ react (StrictMode, createRoot)
├─ ./index.css
└─ ./App.jsx

client/src/App.jsx
├─ react-router-dom (BrowserRouter, Routes, Route, Navigate, Link, useLocation)
├─ ./context/AuthContext (AuthProvider, useAuth)
├─ lucide-react (Home, Lightbulb, PlusCircle, User, Logout)
├─ ./pages/Register
├─ ./pages/Login
├─ ./pages/Home (as HomeFeed)
├─ ./pages/CreateIdea
├─ ./pages/IdeaDetail
└─ ./pages/Profile

client/src/context/AuthContext.jsx
├─ react (createContext, useContext, useState, useEffect)
└─ axios

client/src/pages/Home.jsx
├─ react (useState, useEffect, useCallback)
├─ react-router-dom (Link)
├─ lucide-react (MessageCircle, ThumbsUp, ThumbsDown)
└─ axios

client/src/pages/Login.jsx
├─ react (useState)
├─ react-router-dom (useNavigate, Link)
├─ ../context/AuthContext (useAuth)
└─ lucide-react (Lightbulb)

client/src/pages/Register.jsx
├─ react (useState)
├─ react-router-dom (useNavigate, Link)
├─ ../context/AuthContext (useAuth)
└─ lucide-react (Lightbulb)

client/src/pages/CreateIdea.jsx
├─ react (useState)
├─ react-router-dom (useNavigate)
├─ lucide-react (ArrowLeft)
└─ axios

client/src/pages/IdeaDetail.jsx
├─ react (useState, useEffect)
├─ react-router-dom (useParams, useNavigate)
├─ ../context/AuthContext (useAuth)
├─ lucide-react (ThumbsUp, ThumbsDown, MessageCircle, Trash2, Edit2, ArrowLeft)
└─ axios

client/src/pages/Profile.jsx
├─ react (useState, useEffect)
├─ react-router-dom (useParams, Link)
├─ ../context/AuthContext (useAuth)
├─ lucide-react (ThumbsUp, ThumbsDown, ArrowLeft)
└─ axios
```



## 14. Unused Assets & Notes

### Unused Files

| File                          | Status        | Notes                                                                                     |
|-------------------------------|---------------|-------------------------------------------------------------------------------------------|
| client/src/App.css            | Empty         | All styles consolidated into index.css                                                    |
| client/public/icons.svg       | Unused        | SVG sprite with bluesky, discord, github, x, social, documentation icons — never imported |
| client/src/assets/hero.png    | Unused        | Image file — never imported                                                               |
| client/src/assets/react.svg   | Unused        | React logo — never imported                                                               |
| client/src/assets/vite.svg    | Unused        | Vite logo — never imported                                                                |
| Home.jsx MessageCircle import | Unused import | Imported but never rendered in JSX                                                        |

### Test Coverage

| Test File       | Tests     | Database                 |
|-----------------|-----------|--------------------------|
| health.test.js  | 1         | —                        |
| auth.test.js    | 7         | idea-vault-test-auth     |
| idea.test.js    | 7         | idea-vault-test-ideas    |
| score.test.js   | 5         | — (unit test)            |
| vote.test.js    | 6         | idea-vault-test-votes    |
| comment.test.js | 6         | idea-vault-test-comments |
| <b>Total</b>    | <b>34</b> |                          |

**Note:** Each test file uses its own MongoDB database to avoid interference between parallel test runs.

### Key Design Decisions

- **Score is computed, not stored:** The `computeScore()` function runs at read time on every GET request for ideas.
- **Cookie-based auth:** JWT stored in httpOnly cookies, not localStorage. More secure against XSS.
- **One vote per user per idea:** Enforced by a compound unique index in MongoDB.
- **Author-only controls:** Only the idea author can close/reopen their ideas or delete any comment.
- **Comment edit is per-author:** Users can only edit their own comments; idea authors can delete any comment.